

Mixture viscosity measurement — Solution

W26 (2026) — English translation

A1

0.50 pts

Express the mole fraction of water n_w in solution in terms of the mass fraction of glycerol w_g , μ_w and μ_g .

The amount of a substance in a mixture is defined as:

$$\nu = \frac{m}{\mu},$$

where m is the mass of the substance, and μ is its molar mass. Let us write the formula for the mole fraction of water:

$$n_w = 1 - n_g = 1 - \frac{\nu_g}{\nu_g + \nu_w},$$

from which we obtain:

Answer:

$$n_w = 1 - \frac{w_g \mu_w}{\mu_g - w_g (\mu_g - \mu_w)}$$

A2

2.00 pts

Using the equation (2), convert the values of T_g to the values of w_g .

Using the tabular method or the simple iteration method with $a = 0.9$, we obtain:

Answer:

T_g , K	w_g	η , Pa · s
195.2	1.000	0.906
194.8	0.996	0.839
194.6	0.994	0.767
193.9	0.987	0.655
193.1	0.979	0.558
191.8	0.965	0.479
190.8	0.954	0.373
188.6	0.930	0.202
183.4	0.869	0.0674
173.0	0.728	0.0278
170.0	0.682	0.0134
154.6	0.384	0.0026
149.3	0.252	0.0017
143.8	0.100	0.0012
140.3	0	0.0009

A3

0.30 pts

Propose a linearization that describes a system with a low mole fraction of water, which can be used to determine the parameters ξ and k .

Answer: The dependence of $\ln\left(\frac{\eta}{1 \text{ Pa} \cdot \text{s}}\right)$ on n_w is linear.

A4

1.20 pts

Plot the linearized relationship over the range $n_w \in [0, 0.2]$. Determine the parameters ξ and k using the resulting graph.

Let us recalculate the data to find the dependence of $\ln\left(\frac{\eta}{1 \text{ Pa} \cdot \text{s}}\right)$ on n_w .

$T_g, \text{ K}$	w_g	$\eta, \text{ Pa} \cdot \text{s}$	n_w	$\ln \eta$
195.2	1.000	0.906	0	-0.099
194.8	0.996	0.839	0.020	-0.176
194.6	0.994	0.767	0.030	-0.265
193.9	0.987	0.655	0.063	-0.423
193.1	0.979	0.558	0.099	-0.583
191.8	0.965	0.479	0.156	-0.736
190.8	0.954	0.373	0.198	-0.986
188.6	0.930	0.202	0.278	-1.600
183.4	0.869	0.0674	-	-
173.0	0.728	0.0278	-	-
170.0	0.682	0.0134	-	-
154.6	0.384	0.0026	-	-
149.3	0.252	0.0017	-	-
143.8	0.100	0.0012	-	-
140.3	0	0.0009	-	-

Approximating the dependence with a straight line, we obtain:

$$k = 4.31,$$

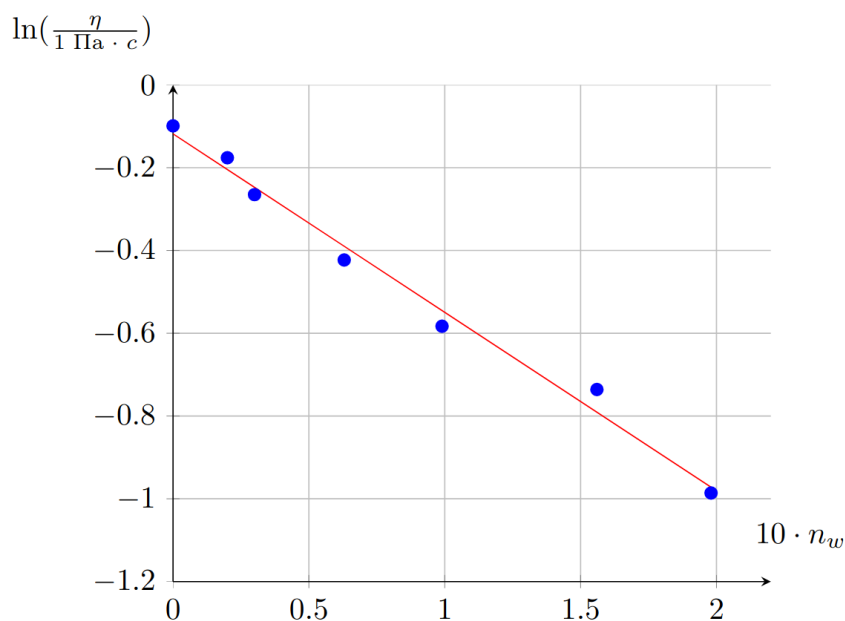
$$\ln\left(\frac{\xi}{1 \text{ Pa} \cdot \text{s}}\right) = -0.12,$$

from which it follows:

Answer:

$$k = 4.31$$

$$\xi = 0.887 \text{ Pa} \cdot \text{s}$$



A5

0.30 pts

Knowing the parameters ξ and k , propose a linearization that allows you to determine the values of the parameters a and b .

Answer: The dependence of $\ln \left| \frac{\xi \exp(-kn_w)}{\eta} - 1 \right|$ on n_w is linear.

A6

1.20 pts

Plot the linearized relationship. From the graph, determine a and b . Note: due to the inaccuracy of experimental data, some values cannot be recalculated. There are no more than 5 such values.

Note: due to the inaccuracy of experimental data, some values cannot be recalculated. There are no more than 5 such values.

Let us continue recalculating the data points:

T_g , K	w_g	η , Pa · s	n_w	$\ln \left \frac{\xi \exp(-kn_w)}{\eta} - 1 \right $
195.2	1.000	0.906	0	-3.92
194.8	0.996	0.839	0.020	-3.54
194.6	0.994	0.767	0.030	-4.12
193.9	0.987	0.655	0.063	-3.44
193.1	0.979	0.558	0.099	-3.28
191.8	0.965	0.479	0.156	-2.91
190.8	0.954	0.373	0.198	-4.35
188.6	0.930	0.202	0.278	-1.12
183.4	0.869	0.0674	0.435	0.02
173.0	0.728	0.0278	0.656	-0.12
170.0	0.682	0.0134	0.704	0.78
154.6	0.384	0.0026	0.891	1.85
149.3	0.252	0.0017	0.938	2.10
143.8	0.100	0.0012	0.979	2.29
140.3	0	0.0009	1	2.50

Let us plot the resulting points on the graph: $\ln \left| \frac{\xi \exp(-kn_w)}{\eta} - 1 \right|$ versus n_w .

In the range $n_w \in [0, 0.2]$, the expression inside the logarithm changes sign and is close to zero, so the function behaves non-monotonically in this region. To perform a linear fit, we consider only points in the range $n_w \in [0.7, 1]$.

For $n_w > 0.7$, $\frac{\xi \exp(-kn_w)}{\eta} > 1$ holds true, so $a > 0$. By drawing the approximating straight line, we find:

$$b = 5.66,$$

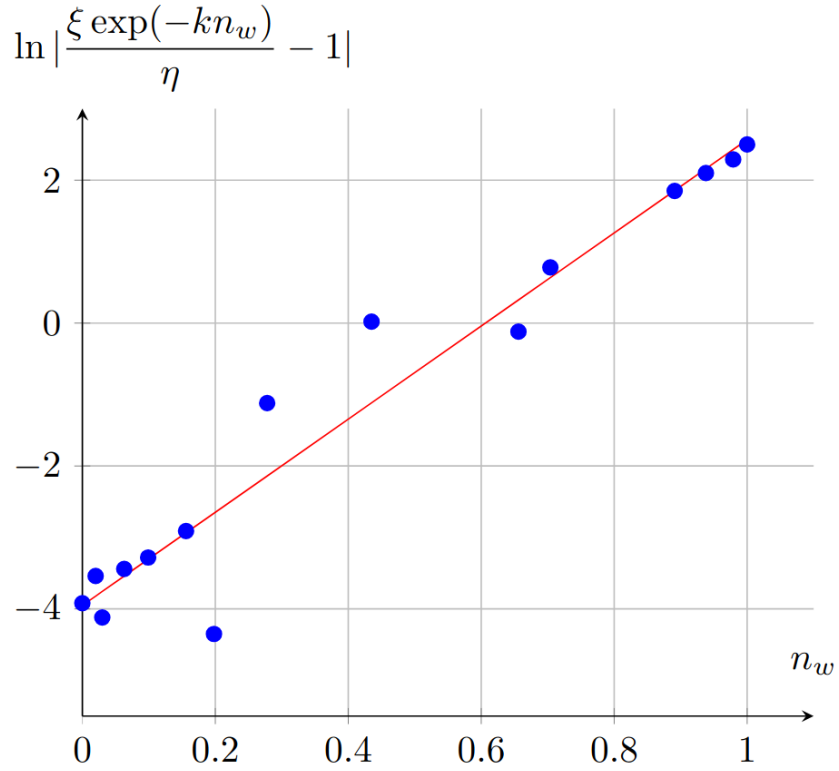
$$\ln a = -3.21,$$

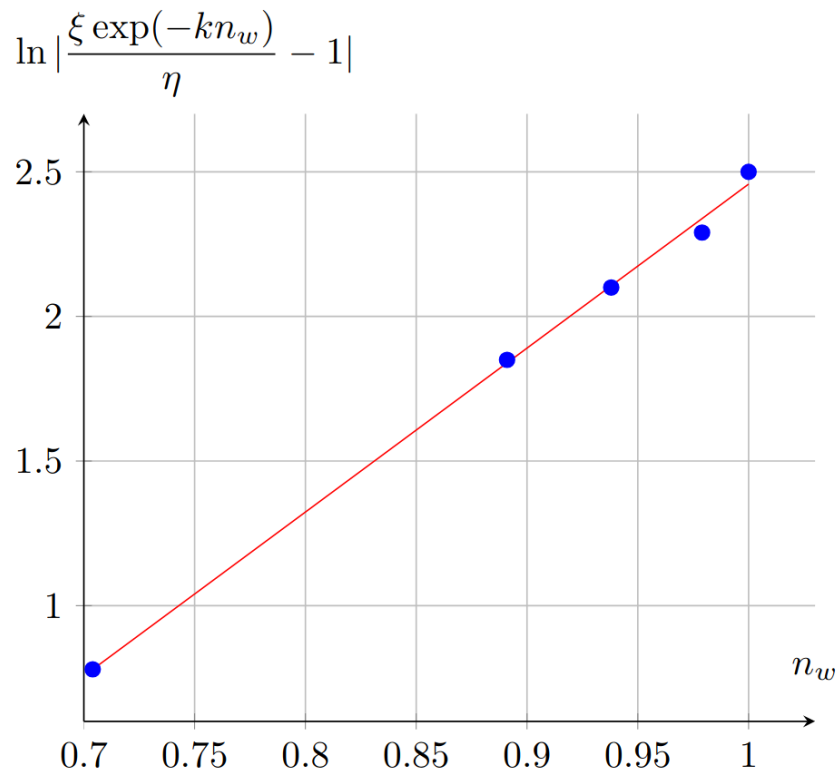
from which we obtain:

Answer:

$$a = 0.040$$

$$b = 5.66$$





A7

1.00 pts

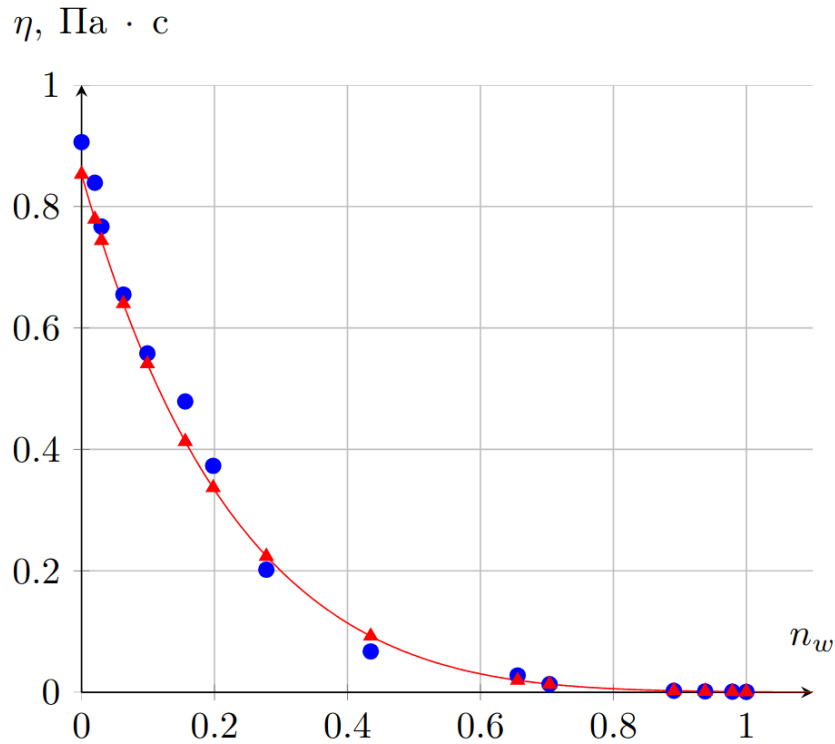
Draw a graph of the theoretical and experimental dependencies $\eta(n_w)$ on one piece of graph paper.

Let us calculate the theoretical values for $\eta(n_w)$:

$T_g, \text{ K}$	w_g	$\eta, \text{ Pa} \cdot \text{ s}$	n_w	$\ln \eta$	$\ln \left \frac{\xi \exp(-kn_w)}{\eta} - 1 \right $	$\eta_{\text{th}}, \text{ Pa} \cdot \text{ s}$
195.2	1.000	0.906	0	-0.099	-3.92	0.853
194.8	0.996	0.839	0.020	-0.176	-3.54	0.779
194.6	0.994	0.767	0.030	-0.265	-4.12	0.744
193.9	0.987	0.655	0.063	-0.423	-3.44	0.640
193.1	0.979	0.558	0.099	-0.583	-3.28	0.541
191.8	0.965	0.479	0.156	-0.736	-2.91	0.413
190.8	0.954	0.373	0.198	-0.986	-4.35	0.337
188.6	0.930	0.202	0.278	-1.600	-1.12	0.224
183.4	0.869	0.0674	0.435	-	0.02	0.0926
173.0	0.728	0.0278	0.656	-	-0.12	0.0199
170.0	0.682	0.0134	0.704	-	0.78	0.0135
154.6	0.384	0.0026	0.891	-	1.85	0.0026
149.3	0.252	0.0017	0.938	-	2.10	0.0017
143.8	0.100	0.0012	0.979	-	2.29	0.0012
140.3	0	0.0009	1	-	2.50	0.0010

Let us build a graph of the obtained dependencies: $\eta(n_w)$.

By drawing smoothing curves, we conclude: the equation is applicable over the entire range ($n_w \in [0, 1]$).



B1

0.50 pts

Using the equation (2), determine the glass transition temperature of the solution T_g .

Let us express w_g :

$$w_g = \frac{m_g}{m_g + m_w} = \frac{\rho_g c_g}{(\rho_g - \rho_w)c_g + \rho_w},$$

then we have:

Answer:

$$T_g = \left[140.3 + 35.272 \frac{\rho_g c_g}{(\rho_g - \rho_w)c_g + \rho_w} - 3.879 \left(\frac{\rho_g c_g}{(\rho_g - \rho_w)c_g + \rho_w} \right)^2 + 23.467 \left(\frac{\rho_g c_g}{(\rho_g - \rho_w)c_g + \rho_w} \right)^3 \right] \text{ K} \approx 162.8 \text{ K}$$

B2

2.00 pts

Propose a linearization to determine the values of the parameters D and T_0 . After plotting the linearized dependence, determine the values of these parameters.

The dependence of $\frac{1}{\ln(\eta/\eta_0)}$ on T is linear with a slope $k = \frac{1}{DT_0}$ and a y-intercept $b = -\frac{1}{D}$. Let us recalculate the data points using this formula and plot the linearized dependence:

$T, ^\circ\text{C}$	$\eta, \text{Pa} \cdot \text{s}$	T, K	$\frac{1}{\ln(\eta/\eta_0)}$
1	0.0201	274	0.556
4	0.0173	277	0.607
11	0.0128	284	0.743
15	0.0104	288	0.878
35	0.0046	308	3.10
44	0.0034	317	48.1
50	0.0031	323	-14.0
56	0.0026	329	-4.04
60	0.0022	333	-2.41
67	0.0020	340	-1.96
71	0.0017	344	-1.487
76	0.00145	349	-1.20
85	0.0013	358	-1.06
90	0.0011	363	-0.903
96	0.0010	369	-0.831
100	0.0009	373	-0.764

From the proposed equation, it follows that it is only valid at sufficiently low temperatures. It is clear from the graph that the equation is applicable for the first 4 points of the graph (at $T < 288 \text{ K}$).

Having determined the parameters of the approximating straight line, we obtain a system of equations:

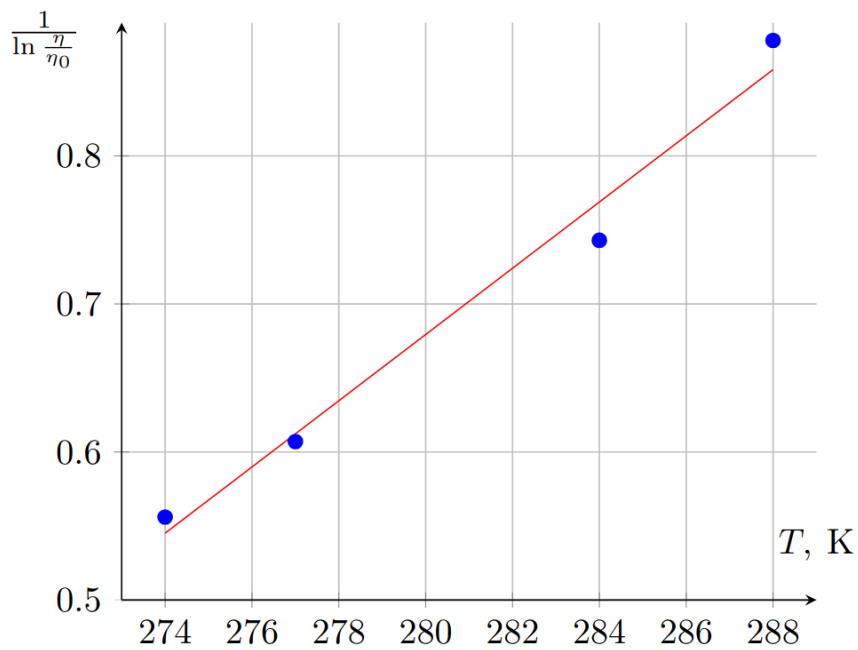
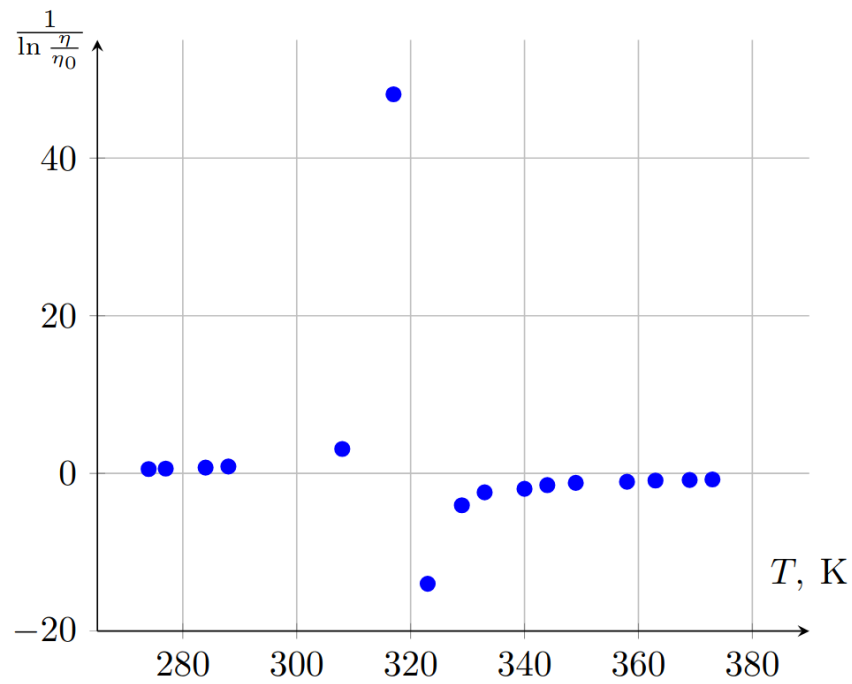
$$\begin{cases} -\frac{1}{P} = -5.60 \\ \frac{1}{DT_0} = 0.022 \text{ K}^{-1}, \end{cases}$$

from which it follows:

Answer:

$$T_0 = 255 \text{ K}$$

$$D = 0.18$$



C1

0.40 pts

Determine the viscosity of the mixture η_n at the volume fraction of glycerol $c_g = 0.5$ and temperature $T = 25^\circ\text{C}$ using the dependence on the mole fraction of water.

The equation obtained for $\eta_n(n_w)$ is applicable over the entire range. Therefore, it is enough to

convert c_g into n_w :

$$n_w = 1 - \frac{w_g \mu_w}{\mu_g - w_g(\mu_g - \mu_w)} = 1 - \frac{\frac{\rho_g c_g}{(\rho_g - \rho_w)c_g + \rho_w} \mu_w}{\mu_g - \frac{\rho_g c_g}{(\rho_g - \rho_w)c_g + \rho_w} (\mu_g - \mu_w)} \approx 0.80$$

and substitute it into the resulting equation. Thus, we obtain:

Answer: $\eta_n = 6.0 \text{ mPa} \cdot \text{s}$

C2

0.40 pts

Determine the viscosity of the mixture η_t under the same conditions using the temperature dependence.

The equation obtained for $\eta_t(T)$ is not directly applicable for $T = 25 \text{ }^\circ\text{C}$. You can determine the viscosity at a given temperature graphically by plotting the curve $\eta_t(T)$, but a result with good accuracy is given by a linear approximation:

$$\eta_{25} \approx \frac{\eta_{15} + \eta_{35}}{2} = 7.5 \text{ mPa} \cdot \text{s}.$$

Answer:

$$\eta_t = 7.5 \text{ mPa} \cdot \text{s}$$

C3

0.20 pts

Compare your results. If there are discrepancies, suggest a possible reason for them.

Answer: The values of η_t and η_n differ slightly. One of the main possible reasons for discrepancies in the results is a poorly measured or fluctuating temperature (for example, if the room temperature changed during the experiment), because the dependence of the mixture's viscosity on temperature is very strong.